Executive Business Report

Sales Forecasting & Demand Intelligence Report
Prepared for: Head of Supply Chain & Chief Financial Officer (CFO)

Executive Summary

This report analyzes the company's historical sales performance and provides a forward-looking view of expected demand over the next three months. The analysis identifies key sales trends, unusual business events, and product demand patterns to support inventory and financial planning. Overall, sales have shown recurring seasonal behavior with clear differences across product categories and regions. Machine learning forecasting predicts relatively stable sales over the next quarter with moderate uncertainty. Product demand segmentation shows that not all products require the same inventory strategy, allowing the business to optimize stock levels while reducing carrying costs. The recommended actions focus on improving inventory availability for high-demand products, reducing excess stock for slow-moving items, and closely monitoring unusual sales spikes that may indicate operational issues or exceptional business events.

Key Findings from Exploratory Data Analysis (EDA)

1. Historical Sales Performance
   - Sales demonstrate recurring monthly seasonality, indicating predictable buying patterns.
   - Certain regions consistently generate higher revenue than others, suggesting geographical demand concentration.
   - A small number of product categories contribute a large proportion of total sales.
   - Monthly sales fluctuate significantly during promotional and holiday periods.
2. Product Performance
   - High-selling sub-categories generate the majority of business revenue.
   - Several low-volume products contribute very little revenue while increasing inventory holding costs.
   - Demand varies considerably between product groups, making a single inventory strategy ineffective.
3. Business Insight

   Sales are not evenly distributed across products or locations. Inventory planning should therefore prioritize high-performing products and regions instead of applying uniform stocking policies.

Three-Month Sales Forecast

The forecasting model predicts that overall business demand will remain relatively stable over the next three months with normal seasonal fluctuations.

Model	Jan 2019	Feb 2019	Mar 2019
SARIMA	60331.79	91458.22	97167.57
Prophet	51083.66	90045.4	89661.19
XGBoost	87667.98	90717.43	89354.17

Plain Language Interpretation

The forecast suggests that sales are expected to remain close to current levels. Actual sales may be slightly higher or lower because customer demand, promotions, holidays, and market conditions cannot be predicted perfectly. The uncertainty increases further into the future, which is normal for business forecasting.

• Top Three Anomalies Detected

1. Exceptional Weekly Sales Spike
   - ❖ Likely Cause: Seasonal promotion, Holiday sales campaign, Bulk corporate purchase
   - ❖ Business Impact: Temporary increase in inventory demand requiring additional stock availability.

2. Sudden Sales Drop
   - ❖ Likely Cause: Product stock-out, Supply chain disruption, Reduced customer demand
   - ❖ Business Impact: Potential revenue loss and customer dissatisfaction.

3. Irregular Sales Volatility
   - ❖ Likely Cause
     - Data recording inconsistencies
     - Unexpected operational events
     - Local market fluctuations
   - ❖ Business Impact: Requires monitoring to distinguish genuine business changes from data quality issues.

• Product Demand Segmentation

Using machine learning clustering, products were grouped according to sales performance and purchasing behavior.

Segment	Characteristics	Recommended Stocking Strategy
High Demand	High sales and frequent orders	Maintain higher safety stock and prioritize replenishment
Medium Demand	Stable and predictable sales	Standard inventory levels with regular review

Low Demand	Low sales and infrequent purchases	Reduce inventory levels and replenish only when necessary

- Business Benefit: Applying different inventory strategies for each demand segment can reduce storage costs while improving product availability for customers.

Business Recommendations

1. **Increase Safety Stock for High-Demand Products**
   Maintain additional inventory for products with consistently high sales so customer demand can be met during peak periods and stock-outs are minimized.

2. **Reduce Inventory for Slow-Moving Products**
   Lower inventory levels for products with low sales to reduce storage costs, avoid excess stock, and improve inventory turnover.

3. **Monitor Forecast Deviations Weekly**
   Compare actual sales with forecasted sales every week to quickly identify unusual demand changes, allowing timely adjustments to inventory and business planning.

Risk / Limitation

The forecasting system is based primarily on historical sales data. It cannot fully anticipate unexpected events such as major promotions, competitor actions, economic changes, supply chain disruptions, or sudden shifts in customer behavior. Therefore, forecasts should be reviewed regularly and combined with business knowledge when making important inventory and financial decisions.